Dark Energy from Dark-Dimension Modes with a Running Ansatz

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Abstract

The cosmological-constant problem, a discrepancy of up to $\sim 10^{120}$ between the QFT vacuum energy and the observed dark-energy density, remains unresolved. Recent DESI data releases show a mild preference for evolving rather than constant dark energy. This paper presents a deliberately narrow phenomenological framework in which dark energy arises from Kaluza-Klein graviton modes in the dark dimension, governing an FLRW cosmology. The absolute scale of the dark-dimension radius today is fixed by matching to the observed dark-energy density. The new content of the framework is a separately stated postulate that the running of the dark-dimension radius tracks a parent-horizon scale set by a slowly accreting higher-dimensional bulk source. We work through the resulting effective dark-energy equation of state in detail and identify a specific, falsifiable correlation between the dark-energy equation-of-state parameter and the running of Newton's constant via the 5D Planck relation. The cross-prediction is shown to be independent of the specific running ansatz, reflecting a deeper structural feature of the dark-dimension scenario. Limitations are stated explicitly: a first-principles bulk-flow computation of the dark-energy density, and a derivation of the running postulate from bulk dynamics, are left for future work. The analysis is performed throughout in the quasi-de Sitter regime.

Introduction

The cosmological-constant problem and the observed acceleration

The cosmological constant problem is one of the most severe fine-tuning issues in physics. Quantum field theory predicts a vacuum energy density that vastly exceeds the observed dark-energy density responsible for cosmic acceleration. Recent baryon-acoustic-oscillation results, when combined with supernova and CMB data, show a mild preference for evolving rather than constant dark energy.

Strategy of this paper

Let us adopt a deliberately narrow and falsifiable strategy:

1. Adopt an FLRW cosmology as the background spacetime for the dark-dimension running ansatz.

2. Use the dark-dimension scenario as the quantitative mechanism for a mesoscopic vacuum energy.

3. Postulate, as the substantive content of the framework, that the running of the dark-dimension radius with cosmic scale factor is set by a parent-horizon scale, while the absolute normalization today is fixed by matching to the observed dark-energy density.

4. Translate the running postulate into an effective dark-energy equation of state.

5. Identify a distinctive cross-prediction relating the dark-energy equation-of-state parameter to the running of Newton's constant via the 5D Planck relation, and show that this cross-prediction is independent of the specific functional form of the running.

I do not adopt the upward extension of the cosmological-coupling proposal. The framework presented below requires only a slowly evolving parent geometry, which can be produced by ordinary accretion in the higher-dimensional spacetime; no specific interior metric is required.

Theoretical Foundations

The dark dimension

The dark-dimension proposal posits a single compact extra dimension of mesoscopic size. KK gravitons populating the resulting tower produce, via UV/IR mixing, an effective vacuum energy density which, in natural units, takes the form

$$\rho_{\rm DE} \simeq \frac{\alpha}{R_{\rm DD}^{p}}, \qquad p \simeq 4. \tag{1}$$

The power $p = 4$ corresponds to the optimistic end of the swampland-derived range, and we adopt it throughout. Let us note that the matching coefficient absorbs differences in p for the present-day absolute scale, but the cross-prediction slope derived below depends linearly on p; this dependence is reinstated explicitly where relevant. The framework is constrained by tabletop tests of the inverse-square law.

Higher-dimensional context

The framework sits within the broader landscape of higher-dimensional gravity, in which IR modifications to gravitational dynamics arise generically from extra-dimensional physics (e.g. DGP-type braneworlds). We do not import any specific such mechanism; we use only the dark-dimension parametric scaling (1) together with the running postulate stated below.

Phenomenological Model and Toy Calculation

Cutoff identification: absolute scale and the running postulate

Absolute scale. As in the original dark-dimension scenario, the absolute scale of the dark-dimension radius today is fixed by demanding that the vacuum-energy formula (1) reproduce the observed dark-energy density. The consistency condition involves only the present Hubble scale and the Planck length; the parent-horizon scale does not appear.

Running postulate. The substantive content of the framework is a separately stated postulate: the time evolution of the dark-dimension radius is locked to a slowly-evolving parent-horizon scale through

$$R_{\rm DD}(T) = R_{\rm DD,0}\left(\frac{R_{\rm parent}(T)}{R_{\rm parent,0}}\right)^{n}, \tag{2}$$

where the exponent n parametrizes the ansatz (the original $\sqrt{\cdot}$ form corresponds to $n = 1/2$). Differentiating gives

$$\frac{\mathrm{d}\ln R_{\rm DD}}{\mathrm{d}\ln a} = n\frac{\mathrm{d}\ln M}{\mathrm{d}\ln a},$$

where $M \propto R_{\rm parent}$ is the parent-mass parameter. The parent-horizon framing should be understood as physical motivation; for the dynamical predictions below, only the combination $n\,(d\ln M/d\ln a) = d\ln R_{\rm DD}/d\ln a$ enters, so any mechanism producing the same evolution of $R_{\rm DD}$ would yield the same predictions.

Effective equation of state

We adopt the linear growth ansatz

$$M(T) = M_0\big(1 + \beta T/T_0\big), \qquad |\beta| \ll 1, \qquad T_0 \equiv H_0^{-1}.$$

We work in the quasi-de Sitter regime, in which the late-time expansion is approximately exponential, so that $d\ln a/dT \approx H_0$. This is an idealization at the present epoch ($\Omega_{\rm DE,0} \approx 0.7$, not unity); corrections are subleading in the small parameter β and do not alter the parametric results.

The dark-energy density scales as $\rho_{\rm DE} \propto M^{-pn}$, so

$$\frac{d\ln\rho_{\rm DE}}{d\ln a} = -pn\,\frac{d\ln M}{d\ln a} = -pn\beta.$$

Treating dark energy as an effective fluid (an operational definition; see remark below), the continuity equation gives

$$1 + w = -\frac{1}{3}\frac{d\ln\rho_{\rm DE}}{d\ln a} = \frac{pn\beta}{3} = \frac{4n\beta}{3}.$$

The original $n = 1/2$ recovers $2\beta/3$. This is independent of scale factor; in CPL parametrization it is a constant w_0 with vanishing w_a.

For the quadratic extension $M(T) = M_0[1 + \beta(T/T_0) + \gamma(T/T_0)^2]$, one finds (using $T/T_0 \approx \ln(a/a_0)$ in the quasi-de Sitter regime) $d\ln M/d\ln a \approx \beta + 2\gamma\ln(a/a_0)$, so

$$1 + w(a) \approx \tfrac{4n}{3}\big[\beta + 2\gamma\ln(a/a_0)\big],$$

yielding to leading order $w_0 = -1 + 4n\beta/3$ and $w_a = -8n\gamma/3$, demonstrating that the same logic produces both w_0 and w_a.

Remark on the continuity equation. Because G runs in this framework, the energy–momentum balance is strictly that of a Brans–Dicke-like scalar-tensor theory, in which a kinetic contribution from the modulus enters the Friedmann equation. The expression $1+w = -\frac{1}{3}d\ln\rho_{\rm DE}/d\ln a$ is then to be understood as the *effective* equation of state inferred by an observer fitting the dark-energy density to a power-law evolution at fixed G. For the slow-running regime $|d\ln G/d\ln a| \ll 1$ this matches the leading behavior; a fully self-consistent scalar-tensor analysis is left for future work.

Cross-prediction with the running of Newton's constant

From the 5D Planck relation $M_{\rm Pl}^2 = M_5^3 R_{\rm DD}$ (with the fundamental scale M_5 held fixed), Newton's constant satisfies $G \propto 1/R_{\rm DD}$, so

$$\frac{\mathrm{d}\ln G}{\mathrm{d}\ln a} = -\frac{\mathrm{d}\ln R_{\rm DD}}{\mathrm{d}\ln a} = -n\frac{\mathrm{d}\ln M}{\mathrm{d}\ln a}.$$

Combining with $1 + w = (pn/3)\, d\ln M/d\ln a$, the factor $n\,(d\ln M/d\ln a)$ cancels:

$$\boxed{1 + w_0 \simeq -\frac{p}{3}\frac{\mathrm{d}\ln G}{\mathrm{d}\ln a}\bigg|_0 = -\frac{4}{3}\frac{\mathrm{d}\ln G}{\mathrm{d}\ln a}\bigg|_0}\,.$$

This is a straight line through the origin with slope $-p/3 = -4/3$ (when plotting $1 + w_0$ against $d\ln G/d\ln a$).

Why the result is independent of n. Both $1+w$ and $d\ln G/d\ln a$ depend on the running ansatz only through $d\ln R_{\rm DD}/d\ln a$, since $\rho_{\rm DE}$ and G are each fixed power-law functions of $R_{\rm DD}$ alone. The running rate cancels in the ratio. The cross-prediction therefore tests the *dark-dimension scaling relations themselves* ($\rho_{\rm DE} \propto R_{\rm DD}^{-p}$ and $G \propto R_{\rm DD}^{-1}$), not the specific running mechanism. The slope is sensitive to the dark-dimension power p but not to the running exponent n or the parent-mass growth rate β.

A violation of this relation is therefore a violation of the dark-dimension scaling relations, not merely of the running ansatz.

Observational Consistency and a Distinctive Cross-Prediction

Background-level consistency

The model reduces to a constant-w dark-energy cosmology for linear parent-mass growth and to ΛCDM in the limit of vanishing growth rate. For moderate growth rates the predicted deviation from $w = -1$ is consistent in sign with the mild preference shown by current data. Reproducing both the present-day value and its possible evolution requires the nonlinear extension of the parent accretion history.

A distinctive cross-prediction and an existing tension

The framework predicts a definite line in the plane of dark-energy equation-of-state parameter versus running of Newton's constant. For the DESI-preferred scale $1 + w_0 \sim 0.05$, the cross-prediction implies $|\dot{G}/G| \sim (3/4)(1 + w_0)H_0 \sim 4 \times 10^{-12}\,\mathrm{yr}^{-1}$, an order-of-magnitude tension with the lunar-laser-ranging bound $\sim 10^{-13}\,\mathrm{yr}^{-1}$. I do not invoke an unmotivated screening mechanism to alleviate this tension. The only way the relation survives is if the locally-measured Newton constant is decoupled from the cosmologically-evolving bulk modulus. As things stand, the cross-prediction is a strong, sharp, currently-tensioned prediction. Let us regard this as a feature: the framework is testable now.

Auxiliary predictions

KK graviton modes are expected to contribute to the stochastic gravitational-wave background; this signature is inherited from the dark-dimension scenario. If the higher-dimensional bulk source carries net angular momentum, a small large-scale anisotropy could in principle be imprinted on the cosmology, though the geometric mechanism is less direct than in models with an explicit parent black hole and no quantitative prediction is made here.

Falsifiability

The framework is falsified if future data converge on a $(1 + w_0,\, d\ln G/d\ln a)$ pair inconsistent with the predicted slope, if sub-millimetre gravity tests rule out the required dark-dimension window, or if the predicted relation between equation-of-state parameters is violated.

Final Thoughts

Trading the enormous vacuum-energy fine tuning for a single mesoscopic geometric scale is a feature of the dark-dimension scenario itself. What the framework adds is a phenomenological postulate that ties the running of the dark-dimension radius to a parent-horizon scale, providing a physical origin for the time evolution of dark energy, together with the resulting cross-prediction between dark-energy parameters and the running of Newton's constant. The robustness of the cross-prediction's slope — independent of the running exponent — is a strengthening rather than a weakening, since it elevates the prediction from a test of the embedding to a test of the dark-dimension scaling relations themselves.

Although the focus is late-time dark energy, the synthesis admits one qualitative early-universe imprint worth investigating in future work: if the running postulate holds throughout cosmic history then the effective KK threshold was correspondingly higher in the early universe.

Limitations

The framework is phenomenological. The principal omissions are a derivation of the running postulate from explicit bulk dynamics, a derivation of the dark-energy density from explicit bulk stress-energy flow, a first-principles computation of the accretion parameters, and a quantitative model of the parent-horizon evolution. A self-consistent scalar-tensor treatment of the dark-energy continuity equation in the presence of running G is also outstanding. Two further substantive limitations should be flagged. First, the requirement that the parent-horizon scale evolve slowly enough to match the observed dark-energy density relocates part of the original fine-tuning question to a model parameter. Second, the cross-prediction is at present in significant tension with existing bounds unless local and cosmological moduli decouple. Let us regard the resulting falsifiability as a feature.

- Derivation of the running postulate from explicit higher-dimensional bulk dynamics, including a stabilization mechanism.
- Quantitative model of the accreting bulk source that sets the parent-horizon scale.
- Self-consistent scalar-tensor analysis of the cosmological dynamics with running G.
- Explicit computation of KK graviton leakage into the bulk.
- Quantitative estimate of the stochastic gravitational-wave background.
- Forecast of the constraining power of future surveys plus improved bounds on $\dot{G}/G$.

- Investigation of mechanisms that could decouple laboratory measurements from the cosmologically-evolving bulk modulus.

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